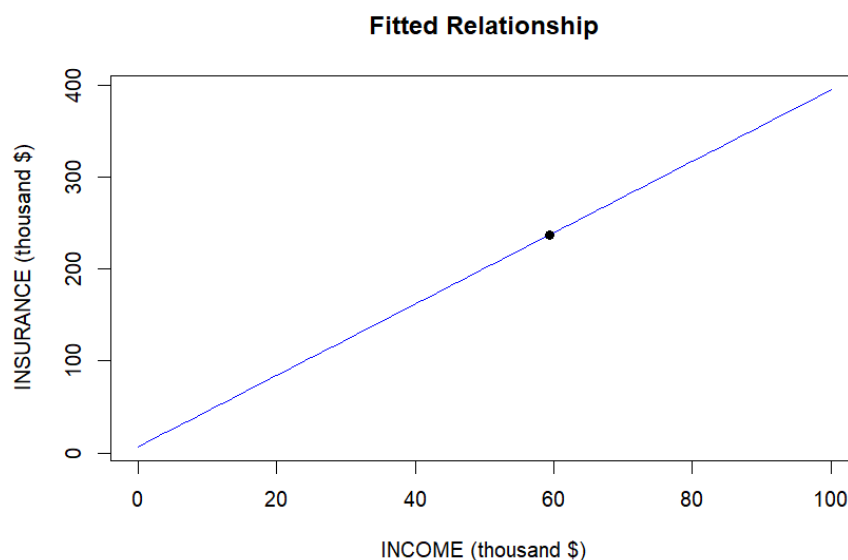


A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N=20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.



sample mean of the amount of insurance held $= 6.855 + 3.880 \times 59.3 = 236.939$

b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

As the slope of the regression line is 3.880, the average amount of insurance held will increase $3.880 \times 1,000 = 3,880$ with each additional 1,000 of household income.

The 95% confidence interval is $[3.645, 4.115]$ ($CI = b_1 \pm se(b_1)$). This means that we are 95% confident that the true increase in insurance coverage associated with an additional \$1000 of income lies between \$3,645 and \$4,115.

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .

For a household with income of \$100,000, the predicted amount of insurance is $\hat{y} = 6.855 + 3.880 \times 100 = 394.855$ (thousand dollars).

The variance of the predicted value is 5.545. ($Var(\hat{y}) = Var(b_0) + x_0^2 Var(b_1) + 2x_0 Cov(b_0, b_1)$)
The 99% confidence interval is $[378.8936, 410.8164]$. Thus, we are 99% confident that the expected amount of insurance held by a household with \$100,000 income lies between \$378,894 and \$410,816.

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $\text{INSURANCE} = \beta_1 + \beta_2 \text{INCOME} + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

The claim implies that the slope coefficient is equal to 5. Thus, the hypotheses are:

$$H_0: \beta_2 = 5$$

$$H_1: \beta_2 \neq 5$$

The test statistic is

$$t = \frac{\hat{\beta}_2 - 5}{se(\hat{\beta}_2)} = \frac{3.880 - 5}{0.112} = -10.00$$

At the 5% significance level, the critical value is approximately 2.100922. The rejection region is: $|t| > 2.101$. Since $|-10.00| = 10.00 > 2.101$, we reject the null hypothesis.

Therefore, the data do not support the management board's claim that the average amount of insurance increases by \$5000 for every \$1000 increase in income.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

The claim implies that the slope coefficient is larger than 1. Thus, the hypotheses are:

$$H_0: \beta_2 = 1$$

$$H_1: \beta_2 > 1$$

The test statistic is

$$t = \frac{\hat{\beta}_2 - 1}{se(\hat{\beta}_2)} = \frac{3.880 - 1}{0.112} \cong 25.71429$$

At the 1% significance level, the critical value is approximately 2.55238. The rejection region is: $t > 2.55238$. Since $25.71429 > 2.55238$, we reject the null hypothesis.

Thus, there is strong evidence that the slope is greater than one, meaning that the amount of life insurance increases by more than the increase in income.